

Lecture Notes on

Unit 02 Differential Equations

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Within the contents of this is my study guide/ personal notes to MAP2302 which was taken in the Spring of 2026 at Palm Beach State (PBSD) Community College in Lake Worth, Florida under Professor Tamara Johns.

This chapter predominantly covers the following topics (General):

1. Lesson 7 Higher-Order Linear Differential Equations
2. Lesson 8 Higher-Order Differential Equations / Reduction of Order
3. Lesson 9 Homogenous Linear Equations with Constant Coefficients
4. Lesson 10 Undetermined Coefficients - Superposition Approach
5. Lesson 11 Undetermined Coefficients - Annihilator Approach

The main revisions made in this study guide versus Unit 01 will be the organization and the utilization of some sort of Appendix to be able to keep track of misc things/ concepts (int. by parts shortcuts to save time, intermediate trig, etc.)

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1 Higher Order Linear Differential Equations

1.1 Higher Order Linear ODEs (introduction)

1.2 Higher Order Linear ODEs (continued)

1.3 Second Order Linear ODEs (introduction)

1.4 How to Compute a Wronskian & Linear Independence

2 Higher Order Linear Diff. Eq./ Reduction Of Order

2.1 Reduction of Order (introduction)

2.2 Reduct. of Order (Additional Homogeneous Equations Examples)

2.3 Reduct. of Order (Second-Order Non-Homogeneous Exmaples)

3 Homogeneous Linear Equations with Constant Coefficients

3.1 Second-Order Homogeneous Equations (Constant Coefficients Introduction)

3.2 Second-Order Equations with Constant Coefficients (Distinct Real Roots)

3.3 Second-Order Equations with Constant Coefficients (Repeated Real Roots)

3.4 Second-Order Equations with Constant Coefficients (Complex Roots)

4 Undetermined Coefficients - Superposition Approach

4.1 Higher order linear ODEs

4.2 Nonhomogeneous equations

4.3 Method of Undetermined Coefficients (Second Order Non-Homogeneous Equations)

4.4 Undetermined Coefficients (Multiplying by x for Linear Independence)

5 Undetermined Coefficients - Annihilator Approach

5.1 How to use the Annihilator Method to Solve a Differential Equation

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